27/7/26

Experiment-2

Aim:

Study of different types of Network Cables.

a) Understand different Types of network cable

⇒ Different type of cables used in networking are

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. Co-axial cable
4. Fibre optic cable

Cable Type	Category	Maximum Data Transmission.	Advantages / Distanvantages	Application / use
UTP	category 3	10 bps	Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter	10 Base-T ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	category 6, 6a	10 Gbps	Adv: • Shielded • Faster than UTP • Less susceptible to noise + interface Disadv: • Expensive • Greater Installation effort	Gigabit Ethernet 10 Gb Ethernet (55m) Widely used in data centers.
SSTP	category 7	10 Gbps		Gigabit Ethernet, 10 Gb Ethernet (100m).

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application /use
Coaxial Cable	RG-6 RG-59 RG-11	10-100Mbs	Adv: • High bandwidth • Immune to interference • Low Loss bandwidth • versatile • Disadv: • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television Network High speed Internet connections.
Fibre optic cable	Single mode Multi mode	100 Gpbs	Adv: • High speed • High bandwidth • High security • Long distance Disadv: • Expensive • Requires skilled Installers	• Maximum distance of fibre optics cable is around 100 meters.

b) Make Your Own Ethernet Cross-Over Cable/Straight Cable

Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the sheilding off the cables.
- Two RJ45 plugs. • Optional two plug shields.

STEPS

Step 1: To start construction of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this Tasks

Step 3: After, you will need to untangle the wires: there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other B.

Step 4: Once the order is correct, bunch them together in a line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order.

Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding. and if it isn't, that means that you stripped off too much shielding.

Step 6: After the wires are securly sitting inside the plug, insert it into the crimping tool and push down.

Step 7: Lastly, repeat for the other end using diagram B to make cross over cables) using diagram A.

Result:

Thus the attempt to connect two devices directly has been done success fully.